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Requirement Analysis and Specification Document

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1 Introduction

1.1 Purpose

BBP is a software designed to simplify the life of everyone who goes on bike trips and wants to enjoy them without worrying about finding the best routes.

It is intended as a community for cyclists, in which every person can share information about their personal routes, so that everyone may take advantage of the information and experience of other bikers. The application is intended for smartphone, where it can make use of GPS and sensor data to seamlessly reconstruct the path a cyclist has taken.

1.1.1 Goals

- G1: Users can record their bike trips through an automated process
- G2: Users can manually insert and record data about their bike trips
- G3: Users can view statistics and status of recorded bike trips
- G4: Users can access, if available, meteorological data about the bike trip
- G5: Users can view potential hazards identified on the bike path
- G6: Users can publish recorded bike trips
- G7: Users can access all published bike trips
- G8: Users can view, on a map, all published bike paths between two points, ordered based on their score

1.2 Scope

1.2.1 World Phenomena

- WP1: User goes on a bike trip
- WP2: User wants to register his bike trips
- WP3: User encounters obstacles
- WP4: User wants to find the best bike paths
- WP5: User wants to access information about bike paths
- WP6: Meteorological conditions change along the bike path
- WP7: User wants to share information about their bike paths

1.2.2 Shared Phenomena

- SP1: User creates an account
- SP2: User logs in the account
- SP3: User records a bike trip
- SP4: Application stores inserted bike trip

- SP5: Application calculates and provides statistics about bike trips
- SP6: App retrieves meteorological informations along the path from an external service
- SP7: App adds meteorological informations to bike path data
- SP8: User insert status and name of the streets in the bike path
- SP9: User insert presence of obstacles along the path
- SP10: User let BBP acquire data from his mobile phone automatically
- SP11: BBP guess if the user is biking given their speed
- SP12: BBP collects GPS information from users phone to reconstruct the followed path
- SP13: BBP acquires data from users phone sensors to identify potential problems on the path
- SP14: User confirms information acquired automatically from BBP
- SP15: User corrects information acquired automatically from BBP
- SP16: User publishes collected information on BBP
- SP17: Any user asks BBP to visualize bike path on a map, specifying origin and destination
- SP18: BBP provides all shared information about selected bike paths to any user
- SP19: BBP computes score of a path based on status and effectiveness on reaching the destination
- SP20: BBP visualizes all paths between the two point inserted by user, ordered based on the computed score

1.3 Definitions, Acronyms, Abbreviations

1.4 Revision History

1.5 Reference Documents

1.6 Document Structure

2 Overall Description

3 Specific Requirements

4 Formal Analysis Using Alloy

5 Effort Spent